Sylow Theorems

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1 Preliminaries

In this section, we provide all the theorems and the definitions that we have studied in Modern Algebra, discrete mathematics courses and we used in the following sections. For the sake of time, we omit the proofs.

Definition 1.1: Equivalence relation

A relation R on a set A is called an *equivalence relation* if it satisfies the following three properties:

- For any $a \in A$, we have that aRa (Reflexivity)
- For any $a, x \in A$, we have that if aRx, then xRa (Symmetry)
- For any $a, x, y \in A$, we have that if aRx and xRy, then aRy (Transitivity)

Definition 1.2: Equivalence class

Let R be an equivalence relation on set A, and let $a \in A$. The equivalence class of a is the set of all elements in A that are equivalent to a. That is, denoted by $[a]_R$ or $[a]$.

$$[a] = \{b \in A \mid b \sim a\}$$

Theorem 1.1

Suppose R is an equivalence relation on a set A. Then we can express A as

$$A = [x_1] \cup [x_2] \cup [x_3] \cup \cdots \cup [x_n]$$

where $x_1, x_2, x_3, \cdots, x_n \in A$ and the equivalence classes $A = [x_1] \cup [x_2] \cup [x_3] \cup \cdots \cup [x_n]$ are pairwise disjoint. In other words, A is the union of the distinct equivalence classes of R

Definition 1.3: Center of Group

The center of a group G is the set

$$Z(G) = \{g \in G \mid \forall h \in G, gh = hg\}.$$

Definition 1.4: Centralizer

The Centralizer of an element $g \in G$ is the set

$$C(g) = \{a \in G \mid ag = ga\}.$$

Theorem 1.2: Lagrange's Theorem

The order of a subgroup of a finite group divides the order of the group.

Definition 1.5: Normal Group

2 Conjugacy Classes

We start by providing the definition of the Conjugacy class, then show that it is a way to partition groups.

Definition 2.1: Conjugacy class

Let a, b be elements of group G. We say that a, b are *conjugate* in G and we say that b is the *conjugate* of a if and only if $gag^{-1} = b$ for some $g \in G$. The set of all conjugates of an element $a \in G$, denoted by $\text{cl}(a) = \{gag^{-1} \mid g \in G\}$, is the ***Conjugacy class of a***.

Theorem 2.1

Conjugacy is an *equivalence relation* on a group.

Proof.

Let R be the conjugacy relation and let G be a group. We show that it is:

- Reflexive: Let $a \in G$. See that $eae^{-1} = a$, so, every element is conjugate to itself and the relation is reflexive.
- Symmetric: Let $a, b \in G$. Suppose that b is conjugate of a, that is, there exists $g \in G$ such that $gag^{-1} = b$. Conjugate b by g^{-1}, $g^{-1}bg = g^{-1}gag^{-1}g = a$. Thus, a is conjugate to b and R is symmetric.
- Transitive: Let $a, b, c \in G$. Suppose that a is conjugate to b, then there exists a $g \in G$ such that $a = gbg^{-1}$, b is conjugate to c, then there exists an $h \in G$ such that $b = hch^{-1}$. Observe that, $a = gbg^{-1} = ghch^{-1}g^{-1} = (gh)c(gh)^{-1}$, and a is conjugate to c. Thus, R is transitive.

■

See that, by Theorem 1.1, we can partition groups into disjoint conjugacy classes.

- **Example**
  In D_4, we have:
  $\text{cl}(R_{90}) = \{R_{90}R_{90}R_{90}^{-1}, R_{180}R_{90}R_{180}^{-1}, R_{270}R_{90}R_{270}^{-1}, R_0R_{90}R_0^{-1},$
  $HR_{90}H^{-1}, VR_{90}V^{-1}, DR_{90}D^{-1}, LR_{90}L^{-1}\} = \{R_{90}, R_{270}\}$

Theorem 2.2

$\forall x \in Z(G)$ we have that $\text{cl}(x) = \{x\}$.

Proof.

Suppose we have a group G, pick an element $a \in Z(G)$. By definition of the center of the group, a commutes with any element in G. Conjugate a by $g \in G$, that is, gag^{-1}. Since a commutes with any element in the group, observe

$$gag^{-1} = agg^{-1} = ae = a.$$

Thus, the conjugacy class of an element in the center of the group is the singleton that contains this element. ∎

The next theorem shows a relationship between the conjugacy class of an element a and the index of the centralizer of a.

Theorem 2.3

Let G be a finite group and let a be an element of G. Then, $|\text{cl}(a)| = |G : C(a)|$.

Proof.

Suppose we have a finite group G and let a be an element of G. To prove that the order of the set of conjugates of an element equals the index of that element centralizer in G we need to find a bijection between the set of conjugates and the set of cosets of the centralizer, let S be the set of left cosets of $C(a)$ in G. Suppose we have the function $\phi : S \to \text{cl}(a)$ such that $\phi(xC(a)) = xax^{-1}, x \in G$.

First, we need to show that ϕ is well defined, that is, if $xC(a) = yC(a)$, then $\phi(xC(a)) = \phi(yC(a))$. Let x, y be elements in the group G and suppose that $xC(a) = yC(a)$. So, $y^{-1}x \in C(a)$ and $x = y(y^{-1}x)$. Hence, since $y^{-1}x$ commutes with a we get that:

$$\phi(xC(a)) = \phi(y(y^{-1}x)C(a)) = y(y^{-1}x)a(y^{-1}x)^{-1}y^{-1} = yay^{-1}.$$

Since we get that $\phi(xC(a)) = \phi(yC(a))$, then the function is well defined.
Next, we show ϕ is injective. Suppose $\phi(xC(a)) = \phi(yC(a))$, then, $xax^{-1} = yay^{-1}$. Multiply both sides by y^{-1} from the left and by x from the right. Observe that $y^{-1}xa = ay^{-1}x$, so $y^{-1}x$ commutes with a and $y^{-1}x \in C(a)$. Since $y^{-1}x \in C(a)$ then, $xC(a) = yC(a)$ as required. Thus, the function ϕ is injective.

Finally, we show that ϕ is surjective. Let b be any element in the conjugacy class $\text{cl}(a)$. By definition, $b = gag^{-1}, g \in G$. We can see the coset $gC(a)$ maps to b. Clearly, the function is surjective.

Therefore, we proved that the function $\phi : S \to \text{cl}(a)$ such that $\phi(xC(a)) = xax^{-1}$ is bijective and the order of the conjugacy class of an element a is the index of the centralizer of that element. ■

Corollary 2.1

Let G be a finite group and let a be an element in G. $|\text{cl}(a)|$ divides $|G|, a \in G$.

Proof.

By the previous theorem, we know that $|\text{cl}(a)| = |G : C(a)|$ We know by Lagrange's Theorem that the index of a subgroup divides the order of the group. Hence, the order of the conjugacy class of a $\text{cl}(a)$ divides the order of the group G. ■

3 The Class Equation

Definition 3.1: The Class Equation

4 The Sylow Theorems

5 Applications of Sylow Theorems